

RENANCE

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# FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MAR 313 — Past Questions & Materials  
**by Renance**

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Course: MAR 313  
Institution: Federal University of Petroleum Resources, Effurun  
Papers in this pack: 3  
Levels: 300

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This study pack was compiled and restructured by Renance for MAR 313 students at Federal University of Petroleum Resources, Effurun. It brings together 3 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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# Paper 1 of 3 — MAR 313 STRENGTH OF MARINE STRUCTURES 2024/2025

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF TECHNOLOGY

DEPARTMENT OF MARINE ENGINEERING

EXAMINATION: First Semester

COURSE CODE: MAR 313

SESSION: 2024/2025

COURSE TITLE: Strength of Marine

Structures

TIME ALLOWED: 2Hre

| Attempt any four (4) questions |

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DATE: 13/05/2025

(Question 1

i. \_ Explain in details the meaning of Hogging and Sagging of Ship Hull Structures. Show Diagrams where applicable. (5 marks)

ii Explain the Causes and the effects of Hogging and Sagging of Ship Hull Structures. Show Diagrams where applicable. (8 marks)

iii. Outline the measures to prevent Hogging and Sagging of Ship Hull Structures. (4.5 marks)

Question 2

i With the aid of illustrative diagrams, Explain the forces acting on a Column. (6 marks)

1s. A Rectangular pontoon barge has a weight of 10 Tonnes/metre; uniformly distributed across its length of 50 meters. If the distances along its length are AB=10 meters, BC=30 meters and CD=10 meters, then a cargo of uniform weight 20 Tonnes/meter was placed at distance BC, Calculate the Bending Moment and Shear forces of the loaded pontoon barge, Sketch the

Diagram

(Calculate the

diagrams as applicable.

(11.5 marks)

(Question 3

A rectangular Pontoon Barge is loaded as shown in the figure below. Estimate the Maximum Bending Moment, given that the length of the barge is 150 meters, light draft is 1 meter. Beam is 30 meters, Depth

is 10 meters and the density of water is 1 tons/meter cube. (17.5 marks)

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Question 4

i Explain the types of Beam studied « (5.5 marks}

is With the aid of Sketf@fies, show the various lypes of structural supports and the Reactions at the supports, (6 marks) |

jis Explain the types of Static Stresses a ship is subsocled to. Show Diavrame wher- splicahble

4) 7) 5

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## **Paper 2 of 3 — MAR 313 STRENGTH OF MARINE STRUCTURES 2023I2024**

Level 300 • First Semester • PastQuestion

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**FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN**  
**COLLEGE OF TECHNOLOGY**  
**DEPARTMENT OF MARINE ENGINEERING**

**EXAMINATION: First Semester**  
**SESSION: 2024/2025**



**COURSE CODE: MAR 313**  
**COURSE TITLE: Strength of Marine Structures**  
**TIME ALLOWED: 2Hrs**

**DATE: 13/05/2025**

Attempt any four (4) questions

**Question 1**

- i. Explain in details the meaning of Hogging and Sagging of Ship Hull Structures. Show Diagrams where applicable. (5 marks)
- ii. Explain the Causes and the effects of Hogging and Sagging of Ship Hull Structures. Show Diagrams where applicable. (8 marks)
- iii. Outline the measures to prevent Hogging and Sagging of Ship Hull Structures. (4.5 marks)

**Question 2**

- i. With the aid of illustrative diagrams, Explain the forces acting on a Column. (6 marks)
- ii. A Rectangular pontoon barge has a weight of 10 Tonnes/meter uniformly distributed across its length of 50 meters. If the distances along its length are AB=10 meters, BC=30 meters and CD=10 meters, then a cargo of uniform weight 20 Tonnes/meter was placed at distance BC. Calculate the Bending Moment and Shear forces of the loaded pontoon barge. Sketch the diagrams as applicable. (11.5 marks)

**Question 3**

A rectangular Pontoon Barge is loaded as shown in the figure below. Estimate the Maximum Bending Moment, given that the length of the barge is 150 meters, light draft is 1 meter, Beam is 30 meters, Depth is 10 meters and the density of water is 1 tons/meter cube. (17.5 marks)

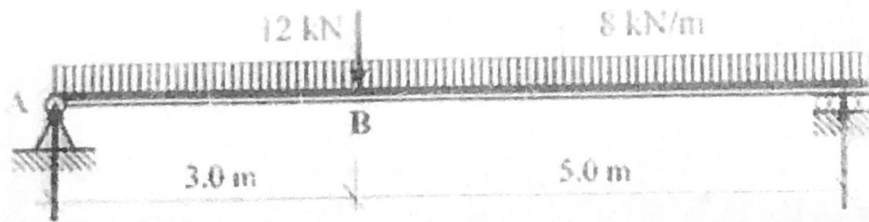
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| 600 T | 900 T | 1500 T | 900 T | 600 T |
| 600 T | 900 T | 1500 T | 900 T | 600 T |

**Question 4**

- i. Explain the types of Beams studied (5.5 marks)
- ii. With the aid of Sketches, show the various types of structural supports and the Reactions at the supports. (6 marks)
- iii. Explain the types of Static Stresses a ship is subjected to. Show Diagrams where applicable. (6 marks)

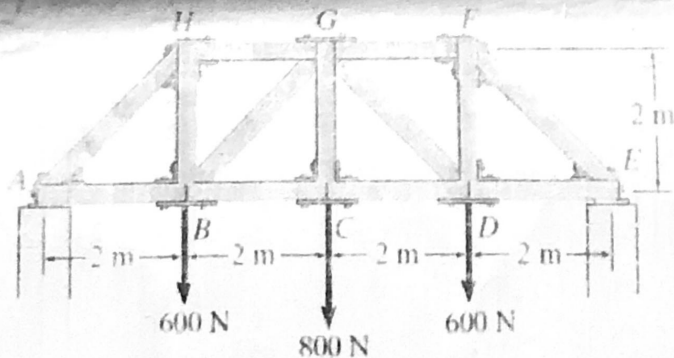
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Question 5

- Briefly explain the basic factors that are considered when designing a structure for Marine applications (6 marks)
- The figure below is a simply supported beam. Calculate the Maximum Shear force and Bending Moment. Also, draw the shear force and Bending Moment Diagrams (11.5 marks)



Question 6

- Discuss the steps to determine the Bending Moment of a ship (7 marks)
- The figure below is a section of the Top side of an Offshore Platform. Determine the forces in members BC, and BG. (10.5 marks)



## Paper 3 of 3 — MAR 313 Strength of Marine 2024/2025.

Level 300 • First Semester • PastQuestion

ge \_ FEDERAL nC UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN  
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i DEPARTMENT OF MARINE ENGINEERING

COURSE CODE: MAR 313

CQURSE TITLE: St rength of Marine

EXAMINATION: First Semester

SESSION: 2024/2025

Structures

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| Attempt any four (4} questions j

Qnestion Y

i. Explain in details the meaning of Hogging and Sagging of Ship Hull Structures. Show Diagrams where applicable. (5 marks)

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iii, | Outline the measures to prevent Hogging and Sagging of Ship Hull Structures. (6. 5 marks)

Onesijon 2

i With the aic of illustrative diagrams, see the forces acting on a Column. (G6 marks)

5. aeciariguier pontoon barge has a weight of 10 Tonnes/meter uniformly distributes across: its length of 50 meters. If the distances along its iength are AB=10 meters, BC=30 meiers and  $\phi$  sD=i $\phi$  meters, then a cargo of uniform weight 20 Tonnvs/rmeter was placed at distance BC, Caleuiz Sending Momeni and Shear forees of the loaded pontoon barge, Ske sich the diagrams gs applicable,

(11.3 marks)

Question 3

A rectangular Pontoon Barge is loaded.as shown in the figure below. Estimate the Moment, given that the length of the barge is 150 meters, light draft is.1 meter, is 10 meters and the density of water is 1-tons'/meter cube. (17.5 marks)

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Maximum Bending

Beam is 30 meters, Depth

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: Explain the types of Beam studied \_

i With the aid of Sketches, show the various types of Supports. (4 marks)

Explain the types of Static Stresses #. ship the subject

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(5.5 marks)

types of structural supports and the Reactions at the

highly ;

Explain 13. Show Diagrams where applicable.

(Explain)

Question 5

Briefly explain the

applications

The figure below is a simply supported beam

Moment, Also, draw the

basic factors

and Bending Moment

section &

Discuss the Steps to determine the Bending Moment

i The figure below is a cross-section of the Top side of a

members. BC, and BG,

that are Considered when designing

Supported beam. Calculate

Shear forces

Offshore Platform. Determine

the Maximum, Shear

moment! Diagrams

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8 Structure for Marine

(6 marks)

and Bending

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